PART III

NON-ELEMENTALISTIC STRUCTURES

The history of human thought may be roughly divided into three periods, each period has gradually evolved from its predecessor. The beginning of one period overlaps the other. As a base for my classification I shall take the relationship between the observer and the observed. . . .

The first period may be called the Greek, or Metaphysical, or Pre-Scientific Period. In this period the observer was everything, the observed did not matter.

The second period may be called the Classical or Semi-Scientific—still reigning in most fields—where the observer was almost nothing and the only thing that mattered was the observed. This tendency gave rise to that which we may call *gross* empiricism and *gross* materialism.

The third period may be called the Mathematical, or Scientific Period. . . . *In this period mankind will understand* (*some understand it already*) *that all that man can know is a joint phenomenon of the observer and the observed.* . . .

Someone may ask, How about "intuitions," "emotions," etc.? The answer is simple and positive. It is a fallacy of the old schools to divide man into parcels, elements; all human faculties consist of an inter-connected whole . . . (280) A. K.

The organism is inexplicable without environment. Every characteristic of it has some relation to environmental factors. And particularly the organism as a whole, *i. e.*, the unity and order, the physiological differences, relations and harmonies between its parts, are entirely meaningless except in relation to an external world. (92) CHARLES M. CHILD

In reality it is the brain as a whole which is the centre of association; and the association is the very raison d'être of the nervous system as a whole. (411) HENRI PIÉRON

The views of space and time which I wish to lay before you have sprung from the soil of experimental physics, and therein lies their strength. They are radical. Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality. (352) H. MINKOWSKI

This assumption is not permissible in atomic physics; the interaction between observer and object causes uncontrollable and large changes in the system being observed, because of the discontinuous changes characteristic of atomic processes. (215) W. HEISENBERG

Well, this is one of the characteristics by which we recognize the facts which yield great results. They are those which allow of these happy innovations of language. The crude fact then is often of no great interest; we may point it out many times without having rendered great services to science. It takes value only when a wiser thinker perceives the relation for which it stands, and symbolizes it by a word. (417) H. POINCARÉ

CHAPTER VIII

GENERAL EPISTEMOLOGICAL

> The physiological gradient is a case of protoplasmic memory since it represents the persistence of the effects of environmental action. The establishment of a gradient in a protoplasm may be regarded as a process of learning.
>
> CHARLES M. CHILD

In what has already been said, we have emphasized repeatedly the 'organism-as-a-whole' principle. The principle is structural, involving most important semantic factors and so deserves a more detailed consideration.

Since the days of Aristotle, more than two thousand years ago, this principle has been often emphasized, often belittled, but, withal, seldom applied. That all we know about life and organisms seems to justify such a principle seems obvious.

The arguments of those experimentalists who belittle or object to such a principle seem to be all of a similar type, and are, perhaps, best expressed by Professor H. S. Jennings, who, in his friendly review of Ritter's book on the *Organismal Conception of Life,* concludes that such an 'organismal conception' is quite justified, but is entirely sterile and does not help laboratory workers.

It must be granted that at the date when the book of Ritter and the review of Jennings were written such a statement was *seemingly* justified. The principle is usually treated as a rough generalization from experience and is not analysed further; the *structural,* epistemological, psycho-logical and semantic consequences were not known, and so the laboratory workers actually did not realize that they *have much help.*

As we have already seen, the main semantic issues were, and are, *structural.* How can we apply the organism-as-a-whole principle if we insist on keeping an *el* language and attitude? Naturally, if the principle is *not* applied, it is futile to look for semantic consequences of a non-applied principle. But once the principle *is* applied, a new language has to be built, of different structure and, *therefore, new implications,* which suggest a long series of new experiments.

A new and structurally different theory may be summarized in a single term—as, for instance, 'tropism' or 'dynamic gradient', a fact which not only revolutionizes our knowledge but which leads also to entirely new experiments and further knowledge. Experiments, as such,

always give relational, structural data, that, under such and such conditions, such and such results follow. The *non-el* attitude and language, as opposed to the old elementalism, is a part of a broader and more fundamental semantic problem; namely, *similarity of structure* between language and the external world. Such similarity leads to similarity of 'logical' relations, predictability, and so forth, and, in general, to the understanding of the structure of the world and *new s.r.*

There are many examples of such organism-as-a-whole terms, but for the present we will mention only the terms 'tropism' in the generalized sense of Loeb, and the 'dynamic or physiological gradients' of Professor Child. The term 'tropism' means the response of the organism-as-a-whole to special external stimuli. For instance, the term 'heliotropism' is applied in cases when the organism responds to the influence of light; 'chemotropism', when it reacts to chemical stimuli; 'galvanotropism', when the organism responds to galvanic (electrical) stimulation ,.

The term 'dynamic or physiological gradient' is the foundation of the $\bar{A}$ biological system of Professor Child. Because of its importance, I shall explain the meaning of this term in some detail.[1]

All protoplasm exhibits empirically a structural characteristic which may be called 'irritability', which appears as a reaction of living protoplasm to external dynamic influences. That 'irritability', as a structural characteristic, becomes obvious when we consider that structurally disintegrated protoplasm is colloidally inactive and becomes 'dead'. Many of the most important characteristics of living protoplasm are strictly bound up with structural integrity.

This 'irritability' occurs in a *structural plenum* and is transmitted to other regions of the protoplasm with differing yet *finite* velocities, and not in 'no time', as Alice would say. Let us imagine a non-differentiated, except for the limiting surface (A), and living bit of protoplasm. This limiting surface represents that part of the protoplasm which is in direct contact with the environment. If the external dynamic factor (S) excites this living bit of protoplasm at a point (B), this stimulus will be the strongest at (B), and it will spread to the further removed portions of (A) in a diminishing gradient. If the decrement is not too sharp, the stimulus will reach the furthest regions of (A); namely, (C), (D), (E), (F),.

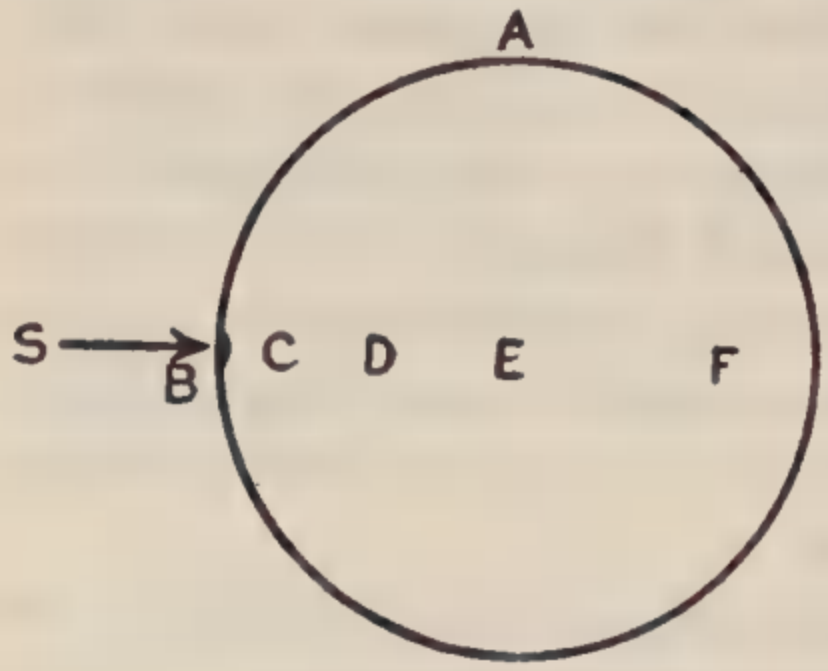

FIG. 1.

The presence or absence of the decrement or its steepness and the intensity of the excitation during transmission depends on the specific character of the protoplasm, and varies from individual to individual, and in different regions and under different conditions varies in one individual.

Thus, we see that a living cell has a necessary relationship with the environment and with external energies because of its limiting surface. The difference between the 'inside of the skin' and the 'outside of the skin' establishes the organism-as-a-whole. The interplay between the inside and the outside is *structural* and supplies the energies which activate the organism. The membrane formation is mostly not dependent upon the constitution of any particular protoplasm, but is rather a general reaction of all protoplasm to environmental influences.

The evidence we have seems to show that in all protoplasm in which we find no specialized conducting paths a certain decrement appears, so that the effectiveness of transmission is limited. In a primitive non-differentiated protoplasm different points further removed from (B) will show different degrees of excitatory changes decreasing from (B). At a certain point the transmission may cease altogether.

The result, then, becomes an excitation-transmission gradient of greater or lesser length, the different levels of which represent various degrees or intensities of excitation.

The primary region of excitation (B) is physiologically more affected and dominant over the other regions to which the excitation is transmitted, because it has more effect upon them than they have upon it. The effect of such conditions gives rise to a temporary structural organismal pattern. The region of primary excitation (B) becomes the dominant region, and the other regions become subordinate to it.

The potentiality for the excitation and the transmission was structurally present in the protoplasm, but this could not produce the pattern which resulted from the external excitation. We see that the action of the external factor was necessary for the realization of the definite physiological pattern whose potentialities were in the protoplasm.

These new excitation-transmission patterns exhibit all the characteristics of new structural patterns in the protoplasmic mass. They determine localized differences at different points, (C), (D), (E),. These differences and *relations* with the dominant region (B) constitute a physiological axis with (B) at one pole. This new pattern constitutes a new structural integration, which is a joint phenomenon of the potentialities of the protoplasm and the environmental action. This relation is of a functional and not merely of a 'plus' character. Child shows that

the physiological axes in their simpler forms are similar to, if not the result of, such excitation-transmission gradients.

For the organism to work as-a-whole, some sort of integrating pattern is necessary. The behaviour of the organism-as-a-whole results, first, from patterns already present, and, second, from the possibilities of further development and integration in response to particular external factors. The physiological gradients give such means.

The development of our nervous system is strictly connected with the above principles discovered by Child.[2] In axiate animals and man the chief aggregation of nervous tissues is localized in the apical (head) end, which region is characterized primarily by a higher rate of metabolism in the early stages. Physiological gradients originate as simple protoplasmic reactions to external stimuli, and so the nervous system originates in protoplasmic behaviour. Nerves then become simply *structuralized* and permanent physiological gradients, and so exert a physiological dominance over other tissues.

From an epistemological point of view, we should notice that the gradients are primarily quantitative and that we do not need specific factors to determine them. Any factor that will determine a more or less persistent quantitative differential in the protoplasm ought to be effective. The above theory is structurally supported by a large number of experiments. For instance, we can experimentally obliterate or determine new gradients.[3] The organism appears in this new light as a behaviour reaction-pattern, and substantiates the old saying that the function builds the organ. Not only should the organism be treated as-a-whole, but it is impossible to isolate the organism from its environment. A functional interrelationship is established between the two.

This theory appears, also, to be fundamental for psychiatry and for psycho-logics, for it establishes the head as a dominant region on the base of an experimentally proven higher rate of metabolism. From Child's point of view, as suggested by Dr. William A. White, the main dynamic gradient, the central nervous axis, gives the *structuralized* evidence of the degree of *correlation* of the other organs and of the degree that the body is under the control of the head-end of this gradient. The failure to keep in touch with this centre of control leads to the disintegration of the individual[4]. The head-end is also the most modifiable point in the axis of control, a conclusion which is of the utmost significance in psychotherapy. It is known that the metabolism of organs can be affected by 'psychic' stimuli, and it is only one step further to understand, as White says, why we may have other structuralized functions, such as structuralized anti-social feelings, structuralized greed, structur-

alized hate., facts which are observed daily in ordinary life and in asylums. From the point of view of the theory of Child, the nervous system appears not only as a structuralized conducting gradient, but it also explains how specific conducting tissues could have evolved from non-specific living protoplasm. It is important to notice the dominance which the primary region of excitation exerts over the others, since, with the great complexities of the human brain, we understand better why so-called 'mental' and semantic issues, which are phylogenetically the youngest, are of such importance.

In our daily life we deal with different people, some of whom are seriously ill 'mentally' and who, under favorable conditions, would be under medical attendance. The majority of us—some specialists consider it to run even as high as ninety per cent of the whole population—would be better off if taken care of by some psychiatrist, or, at least, if under consultation from time to time.

Owing to old religious prejudices, often unconscious, it is still believed that those 'mentally' ill are either obsessed by 'demons' or are being punished for some 'evil',. The majority even of enlightened people have a kind of semantic horror or fright at 'mental' ills, not realizing that under the *animalistic conditions* which prevail at present in our theories, 'ethical', social, economic., those only with the least human traits are favored, while those most human cannot stand such animalistic conditions and often break down. It is not a novelty that a moron cannot be 'insane'. A moron lacks something; only the more gifted individuals, the more human (as compared with animals), break down. I know of many psychiatrists who say that 'it takes a "good mind" to be "insane" '.

Now, 'mental' and semantic excitation, which phylogenetically appeared so recently, naturally plays, in many instances, a dominant part, a fact which science, until very lately, has completely disregarded. The present theory makes it quite obvious that with animalistic theories in existence, and un-sanity (lack of consciousness of abstracting, confusion of orders of abstractions resulting from identifications.,) practically universally operating in every one of us, a seriously unbalanced race must be produced.

There can be no doubt that the consistent application of a *non-el* language in the analysis of animal behaviour has suggested new experiments and that, as a result, the use of such terms had its influence on laboratory workers. It does not matter to what extent these terms, or the theories which they represented, were 'right' or 'wrong', they were terms of the *non-el* type, and they expressed in one term entirely *struc-*

turally new and far-reaching theories. In testing these theories, new series of experiments were required. Even when the new experiments were devised to verify the older experiments, again the laboratory workers got direct benefit of the structurally new terms. But these benefits were largely unconscious, and so biologists *could* believe in the older days that they had no laboratory benefits from the use of such terms; however, this belief is now entirely unjustified.

Since the *non-el* principle is not only a structurally justified empirical generalization, but also involves for its application the structural rebuilding of our language and old theories, the semantic issues are far-reaching and of great practical value.

The application of the principle means the rejection of the old elementalism which results and leads to identifications and to blinding semantic disturbances, which, in turn, prevent clear vision and unbiased creative freedom.

According to the modern theory of materials, as given in Part X, the mutual interdependence, the mutual action and reaction of everything in this world upon everything else appears as a *structural* fact and a necessity, and so *el* languages cannot be expected to lead to satisfactory semantic solutions. We should not be surprised to find that the struggle against identification and elementalism appears at some stage in every science.

Some of the most prominent examples of this tendency outside of biology, psychiatry., can be found in modern physics. From a structural point of view the whole theory of Einstein is nothing else than an attempt to reformulate physics on a *structurally new non-el* and $\bar{A}$ foundation—an exact structural parallel of the biological organism-as-a-whole principle.

Einstein realized that the empirical structure of 'space' and 'time' with which the physicist and the average man deals is such that it cannot be empirically divided, and that we actually deal with a blend which we have split only elementalistically and verbally into these fictitious entities. He decided to build a verbal system closer in structure to the facts of experience and, with the help of the mathematician Minkowski, he formulated a system of new structure which employed a *non-el language* of space-time. As we know from physics and astronomy, this *non-el* language *suggested experiments,* and so it had beneficial laboratory application. But, in fact, the influence goes still deeper, as the present work will show, for such structural advances carry with them profound psycho-logical, semantic effects. Although, at present, these

beneficial influences operate unconsciously, they, nevertheless, tend to counteract the *el* and absolutistic semantic effects of identification.

It is interesting to note that the Einstein theory, because structural, has had the effect upon the younger physicists of a semantic release from the old structural elementalism and has prepared the semantic ground for the crop of young geniuses which has sprung up lately in the quantum field. It was found that the *el* 'absolute' division of the 'observer' and the 'observed' was false to facts, because every observation in this field disturbs the observed. The elimination of this elementalism in the quantum field led to the most revolutionary restricted 'uncertainty principle' of Heisenberg, which, without abolishing determinism, requires the transforming of the two-valued *A* 'logic' into the ∞-valued semantics of probability. Again, this advance in quantum formulations has suggested new experiments.

The $\bar{A}$-system, as originated by the writer in his *Manhood of Humanity* and other writings, is also the result of the structurally *non-el* tendency. In *Manhood of Humanity,* I introduced a *non-el* term, 'time-binding', by which is meant *all* the factors *which as-a-whole* make man a man, and which differentiate him from animals. In carrying the system further in the present book, I reject the structurally *el* separation involved in such terms as 'senses' and 'mind'., and introduce, instead, *non-el* terms, such as 'different orders of abstractions'., where 'mind' and 'senses'., are no longer divided. Curiously enough, even in such a field, the method has suggested experiments, and so again the new language has laboratory importance.

What has been said above about the organism-as-a-whole, and illustrated by particular cases, seems to show a general characteristic of all our abstracting capacities. We usually disregard, or fail to appreciate, the fact that a single structurally important new term might lead to the re-postulation of the whole structure of the language in the given field. In science we search for structure; so any structurally new term is useful, because, when tested, it always gives structural information, whether positive or negative. In our human affairs, it is different. All our human institutions follow the structure of the language used; but we never 'think' of that, and, when the silly institutions do not work, we blame it all on 'human nature', without any scientific justification.

Poincaré, in one of his essays, speaks about the harmful effect which the term 'heat' had on physics. Grammatically, the term 'heat' is classified as a substantive, and so physics was labouring for centuries looking for some 'substance' which would correspond to the substantive name 'heat'. We know by now that there is no such thing, but that 'heat' must

be considered as a manifestation of 'energy'. If we choose to carry this analysis further, we should find that 'energy' is also not a very satisfactory term, but that 'action', perhaps, is more fundamental.

In dealing with ourselves and the world around us, we must take into account the structural fact that everything in this world is strictly interrelated with everything else, and so we must make efforts to discard primitive *el* terms, which imply structurally a *non-existing isolation.*

The moment this is realized, we shall have to treat the *non-el* principle seriously. As the new terms have, also, their *non-el* implications, such terms throw new light on old problems.

In practice, it is difficult, at first, to avoid the use of old terms. When we want to digest fully a new and important work based on new structural terms and acquire corresponding *s.r*, the best way to train oneself in the use of the new terms is by gradually dropping the old terms. If we *have* to use the old terms, then we should train ourselves to be aware of their *insufficiency* and of their *fallacious* structural *implications,* and so be free from the old *s.r.*

The use of the new terms should be deliberate. We should put the problem to ourselves somewhat as follows: The old language is structurally, and, therefore, by implication, semantically unsatisfactory; the new terms seem to correspond closer to facts; let us test the new terms. Are the new terms always structurally satisfactory? Probably not, but in science experiments check predictions, and so new structural issues become clarified.

We have been speaking about new and old terms quite simply, yet the issues are not so simple. The invention of a single structurally new term always involves new structural and relational notions, which, again, involve *s.r.* For instance, if we study any event, and in that study use the terms 'tropism', or 'dynamic gradient', or 'time-binding', or 'order of abstractions', or 'space-time', or 'wave-packets'., we must use all structural and semantic implications the terms involve.

Using the first four terms, we are bound to treat the organism-as-a-whole, for the terms are not *el.* They are not based on the notion of, nor do they postulate, some fictitious 'isolated' elements. In using space-time, we introduce the individuality of events, as every 'point of space' carries with itself a *date,* which, by necessity, makes every 'point' in space-time unique and individual. In using the term 'wave-packet', we re-interpret the older objectified and, perhaps, fictitious 'electron',.

The consistent and permanent use of such terms naturally involves, structurally, a new world-outlook, new *s.r*, more justified by our scientific and daily experience. But the greatest gain is usually in getting

away from primitive structural notions and metaphysics, with their vicious semantic disturbances. In creative work, semantic *limitations* hamper a clear understanding, and prevent scientists from inventing or formulating better, simpler, and more effective theories of different structure.

As soon as we possess 'knowledge', then we shall 'know' all that there is to be known. By definition, there cannot be any *unknowable*. There is a place for the unknown structure. The unknown is rather extensive, partly because science has been, and still is, persecuted, as has already been pointed out.

The so-called 'unknowable' was the semantic result of identification, of a semantic unbalance, which posits for knowledge something 'beyond' knowledge. But has such a postulation any meanings outside of psychopathology? Of course not, as it starts with a self-contradictory assumption, which, being senseless, must lead to senseless results.

We have dwelt on the problems of the structure of terms at such length, because they are generally disregarded, but they are, for semantic purposes, fundamental. The reader will get the main benefit of this book and will receive help in understanding modern scientific issues if he becomes entirely convinced of the seriousness of structural and semantic problems.

Terms are artifices of humans which are necessary to economize effort in the field of 'experience' and experimentation. They are useful in reducing the actual amount of experience necessary, by allowing verbal experimentation. The human rate of progress is swifter than that of the animals, and this is due mainly to the fact that we can summarize and transmit past experiences to the young generation in a degree far more effective than that of the animals. We have also extra-neural means for recording experiences, which the animals lack entirely.

That such verbal experimentation is possible at all is conditioned by the fact that languages have *structure,* and that our knowledge of the world i*s structural* knowledge. Let us repeat once more that if two relations have similar structure, all of their 'logical' characteristics are similar; therefore, once structure is discovered, such a process of verbal experimentation becomes extremely effective, and an accelerating cultural device. The use of an antiquated language in our human affairs, in addition to other drawbacks, prevents our being more intelligent in those affairs.

The natural order of investigation is indicated thus: (1) Empirical search for structure in the sciences; (2) Once this structure is discovered, at each date, the structure of our language is adjusted to it and

our new *s.r* trained. Historically, we have partially followed the reversed, and ultimately pre-human, and so pathological, order. Without science, and with extremely meagre and primitive knowledge of the structure of the world, we have produced grunts and languages of primitive false structure, reflecting, of necessity, its implications as to the assumed structure of the world. We have made out of it primitive dogmas which are still in full sway and embodied in the structure of the old language. This is also the reason why, outside of technical achievements, we are still on such primitive levels. It is easy to understand why experimental science is of such importance and why theoretical (verbal) predictions must be tested experimentally. The above also gives a deeper and a new justification for what is called 'pragmatism'.

Experiments constitute a search for relations and structure in the empirical world. Theories produce languages of some structure. If the two structures are similar, the 'theories work'; otherwise, they do not, and suggest further search and structural adjustments.

It should be mentioned, perhaps, that the main epistemological principle which has led to the writing of the works of the present author was a definite inclination to abandon identification and the resulting structurally unsatisfactory *el* language in general use, and to produce a *non-el* system, which, in structure, would be similar to the world around us, ourselves and our nervous system included. This structural novelty was the foundation on which the $\bar{A}$-system has been gradually built.

CHAPTER IX

COLLOIDAL BEHAVIOUR

> In fact, to-day colloids may be regarded as an important, perhaps the most important connecting link between the organic and the inorganic world. (7)
>
> WOLFGANG PAULI

In our researches, let us follow the natural order and give a brief structural account of what we know, empirically, about the medium in which life is found; namely, about the colloids. The following few elementary particulars show the empirical importance of structure, and so are fundamental in the present work.

At present, physicians are usually too innocent of psychiatry, and psychiatrists, although they often complain about this innocence of their colleagues, seldom, if ever, themselves pay any attention to the colloidal structure of life; and their arguments about the 'body-mind' problem are still scientifically incomplete and unconvincing, though the 'body-mind' problem has been present with us for thousands of years. It is a very important semantic problem, and, as yet, not solved scientifically, although there is a simple solution of it to be found in the *colloidal* structure of life.

The reader should not ascribe any uniqueness of the 'cause-effect' character to the statements which follow, as they may not be true when generalized. Colloidal science is young and little known. Science has accumulated a maze of facts, but we do not have, as yet, a general theory of colloidal behaviour. Statements, therefore, should not be unduly generalized.

We shall only indicate a few structural and relational connections important for our purpose.

When we take a piece of some material and subdivide it into smaller pieces, we cannot carry on this process indefinitely. At some stage of this process the bits become so small that they cannot be seen with the most powerful microscope. At a further stage, we should reach a limit of the subdivision that the particles can undergo without losing their chemical character. Such a limit is called the molecule.* The smallest particle visible in the microscope is still about one thousand times larger than the largest molecule. So we see that between the molecule and the smallest visible particle there is a wide range of sizes.

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*This statement is only approximate, because there is evidence that chemical characteristics change as the molecule is approached.

Findlay calls these the 'twilight zone of matter'; and it was Oswald, I believe, who called it the 'world of neglected dimensions'.

This 'world of neglected dimensions' is of particular interest to us, because in this range of subdivision or smallness we find very peculiar forms of behaviour—life included—which are called 'colloidal behaviour'.

The term 'colloid' was proposed in 1861 by Thomas Graham to describe the distinction between the behaviour of those materials which readily crystallize and diffuse through animal membranes and those which form 'amorphous' or gelatinous masses and do not diffuse readily or at all through animal membranes. Graham called the first class 'crystalloids' and the second 'colloids', from the Greek word for glue.

In the beginning colloids were regarded as special 'substances', but it was found that this point of view was not correct. For instance, NaCl may behave in solution either as a crystalloid or as a colloid; so we began to speak about the *colloidal state*. Of late, even this term became unsatisfactory and is often supplanted by the term *'colloidal behaviour'*.

In general, a colloid may be described as a 'system' consisting of two or more 'phases'. The commonest represent emulsions or suspensions of fine particles in a gaseous, liquid, or other medium, the size of the particles grading from those barely visible microscopically to those of molecular dimensions. These particles may be either homogeneous solids, or liquids, or solutions themselves of a small percentage of the medium in an otherwise homogeneous complex. Such solutions have one characteristic in common; namely, that the suspended materials may remain almost indefinitely in suspension, because the tendency to settle, due to gravity, is counteracted by some other factor tending to keep the particles suspended. In the main, colloidal behaviour is not dependent upon the physical state or chemistry of the finely subdivided materials or of the medium. We find colloidal behaviour exhibited not only by colloidal suspensions and emulsions where solid particles or liquid droplets are in a liquid medium, but also when solid particles are dispersed in gaseous medium (smokes), or liquid droplets in gaseous media (mists),.

Materials which exhibit this special colloidal behaviour are always in a very fine state of subdivision, so that the ratio of *surface exposed* to *volume of material* is very large. A sphere containing only 10 cubic centimetres, if composed of fine particles 0.00000025 cm. in diameter, would have a total area of all the surfaces of the particles nearly equal to half an acre.[1] It is easy to understand that under such *structural* conditions the *surface forces* become important and play a prominent role in colloidal behaviour.

The smaller the colloidal particles, the closer we come to molecular and atomic sizes. Since we know atoms represent electrical *structures*, we should not be surprised to find that, in colloids, surface energies and electrical charges become of fundamental importance, as by necessity all surfaces are made up of electrical charges. The surface energies operating in finely grained and dispersed systems are large, and in their tendency for a minimum, every two particles or drops tend to become one; because, while the mass is not altered by this change, the surface of one larger particle or drop is less than the surface of two smaller ones—an elementary geometrical fact. Electrical charges have the well-known characteristic that like repels like and attracts the unlike. In colloids, the effect of these factors is of a fundamental, yet opposite, character. The surface energies tend to unite the particles, to coagulate, flocculate or precipitate them. In the meanwhile the electrical charges tend to preserve the state of suspension by repelling the particles from each other. On the predominance or intensity of one or other of these factors, the instability or the stability of a suspension depends.

In general, if 'time' limits are not taken into consideration, colloids are *unstable* complexes, in which continuous transformation takes place, which is induced by light, heat, electric fields, electronic discharges, and other forms of energy. These transformations result in a great variation of the characteristics of the system. The dispersed phase alters its characteristics and the system begins to coagulate, reaching a *stable* state when the coagulation is complete. This process of transformation of the characteristics of the system which define the colloid, and which ends in coagulation, is called the 'ageing' of the colloid. With the coagulation complete, the system loses its colloidal behaviour—it is 'dead'. Both of these terms apply to inorganic as well as to organic systems.

Some of the coagulating processes are partial and reversible, and take the form of change in viscosity; some are not. Some are slow; some extremely rapid, particularly when produced by external agencies which alter the colloidal equilibrium.

From what has been said already, it is obvious that colloids, particularly in organisms, are extremely sensitive and complex structures with enormous possibilities as to degree of stability, reversibility., and allow a wide range of variation of behaviour. When we speak of 'chemistry', we are concerned with a science which deals with certain materials which preserve or alter certain of their characteristics. In 'physics', we go beyond the obvious characteristics and try to discover the *structure* underlying these characteristics. Modern researches show clearly that atoms have a very complex structure and that the macroscopic character-

istics are directly connected with sub-microscopic structure. If we can alter this structure, we usually can alter also the chemical or other characteristics. As the processes in colloids are largely structural and physical, anything which tends to have a structural effect usually also disturbs the colloidal equilibrium, and then different macroscopic effects appear. As these changes occur as series of interrelated events, the best way is to consider colloidal behaviour as a physico-electro-chemical occurrence. But once the word 'physical' enters, structural implications are involved. This explains also why all known forms of radiant energy, being structures, can affect or alter colloidal structures, and so have marked effect on colloids.

As all life is found in the colloidal form and has many characteristics found also in inorganic colloids, it appears that colloids supply us with the most important known link between the inorganic and the organic. This fact also suggests entirely new fields for the study of the living cells and of the *optimum conditions for their development, sanity included.*

Many writers are not agreed as to the use of the terms 'film', 'membrane', and the like. Empirically discovered structure shows clearly, however, that we deal with surfaces and *surface energies* and that a 'surface tension film' behaves as a membrane. In the present work, we accept the obvious fact that organized systems are film-partitioned systems.

One of the most baffling problems has been the peculiar periodicity or rhythmicity which we find in life. Lately, Lillie and others have shown that this rhythmicity could not be explained by purely physical nor purely chemical means, but that it is satisfactorily explained when treated as a *physico-electro-chemical structural occurrence.* The famous experiments of Lillie, who used an iron wire immersed in nitric acid and reproduced, experimentally, a beautiful periodicity resembling closely some of the activities of protoplasm and the nervous system, show conclusively that both the living and the non-living systems depend for their rhythmic behaviour on the chemically alterable film, which divides the electrically conducting phases. In the iron wire and nitric acid experiment, the metal and the acid represent the two phases, and between the two there is found a thin film of oxide. In protoplasmic structures, such as a nerve fibre, the internal protoplasm and the surrounding medium are the two phases, separated by a surface film of modified plasm membrane. In both systems, the electromotive characteristics of the surfaces are determined by the character of the film.[2]

That living organisms are film-bounded and partitioned systems accounts also for irritability. It appears that irritability manifests itself as sensitiveness to electrical currents. These currents seem to depend on polarizability or resistance to the passage of ions, owing to the presence of semi-permeable boundary films or surfaces enclosing or partitioning the system. It is obvious that we are here dealing with complex *structures* which are intimately connected with the characteristics of life. Living protoplasm is electrically sensitive only as long as its structure is intact. With death, semi-permeability and polarizability are lost, together with electrical sensitivity.

One of the baffling peculiarities of organisms is the rapidity with which the chemical and metabolic processes spread. Indeed, it is impossible to explain this by the transportation of material. All evidence shows that electrical and, perhaps, other energy factors play an important role; and that this activity again depends on the presence of surfaces of protoplasmic structures with electrode-like characteristics which form circuits.

The great importance of the electrical charges of the colloidal particles arises out of the fact that they prevent particles from coalescing; and when these charges are neutralized, the particles tend to form larger aggregates and settle out of the solution. Because of these charges, when an electrical current is sent through a colloidal solution, the differently charged particles wander to one or the other electrode. This process is called cataphoresis. There is an important difference in behaviour in inorganic and organic colloids under the influence of electrical currents, and this is due to the difference in structure. In inorganic colloids, an electrical current does not coagulate the whole, but only that portion of it in the immediate vicinity of the electrodes. Not so in living protoplasm. Even a weak current usually coagulates the entire protoplasm, because the inter-cellular films probably play the role of electrodes and so the entire protoplasm structurally represents the 'immediate vicinity' of the electrodes. Similarly, structure also accounts for the extremely rapid spread of some effects upon the whole of the organism.

Electrical phenomena in living tissue are mainly of two more or less distinct characters. The first include electromotive energy which produces electrical currents in nerve tissue, the membrane potentials,. The second are called, by Freundlich, electrokinetic, and include cataphoresis, agglutination,. There is much evidence that the mechanical work of the muscles, the secretory action of the glands, and the electrical work of the nerve cells are closely connected with the colloidal structure of these tissues. This would explain why *any factor* (semantic reactions included)

capable of altering the colloidal structure of the living protoplasm must have a marked effect on the behaviour and welfare of the organism.

Experiments show that there are four main factors which are able to disturb the colloidal equilibrium: (1) Physical, as, for instance, X-rays, radium, light, ultra-violet rays, cathode rays.; (2) Mechanical, such as friction, puncture.; (3) Chemical, such as tar, paraffin, arsenic.; and, finally, (4) Biological, such as microbes, parasites, spermatozoa,. *In man, another (fifth) potent factor; namely, the semantic reactions,* enters, but about this factor, I shall speak later.

For our purpose, the effects produced by the physical factors, because obviously structural, are of main interest, and we shall, therefore, summarize some of the experimental structural results. Electrical currents of different strength and duration, as well as acids of different concentration, or addition of metallic salts, which produce marked acidity, usually coagulate the protoplasm, these effects being structurally interrelated. Slow coagulation involves changes in viscosity, all of which, under certain conditions, may be reversible.[3] When cells are active, their fluidity often changes in a sharp and rapid manner.[4]

Fat solvents are called surface-active materials; when diluted, they decrease protoplasmic viscosity; but more concentrated solutions produce increased viscosity or coagulation.[5] The anaesthetics, which always are fat solvents and surface-active materials, are very instructive in their action for our purpose, as they affect very diversified types of protoplasm similarly, this similarity of action being due to the similarity of colloidal structure. Thus, ether of equal concentration will make a man unconscious, will prevent the movement of a fish and the wriggling of a worm, or stop the activity of a plant cell, without permanently injuring the cells.[6] In fact, the action of all drugs is based on their effect upon the colloidal equilibrium, without which action a drug would not be effective. It is well known that various acids or alkalis always change the electrical resistance of the protoplasm.[7]

The working of the organism involves mostly a structural and very important 'vicious circle', which makes the character of colloidal changes *non-additive*. If, for instance, the heart, for any reason, slows down the circulation, this produces an accumulation of carbonic acid in the blood, which again increases the viscosity of the blood and so throws more work on the already weakened heart.[8] Under such structural conditions, the results may accumulate very rapidly, even at a rate which can be expressed as an exponential function of higher degree.

Different regions of the organism have different charges; but, in the main, an injured, or excited, or cooler part is electro-negative (which

is connected with acid formation), and the electro-positive particles rush to those parts and supply the material for whatever physiological need there may be.[9]

The effects of different forms of radiant energy on colloids and protoplasm are being extensively studied, and the results are very startling. The different forms of radiant energy differ in wave-length, frequency.,—that is to say, generally in *structure,*—and, as such, may produce structural effects on colloids and organisms, which effects may appear on the gross macroscopic level in many different forms.

Electrical currents, for instance, retard reversibly the growth of roots, may activate some eggs into larval stages without fertilization., which makes it possible to understand why, in some cases, a mere puncturing of the egg may disturb the equilibrium and produce the effects of fertilization.[10]

The X-, or Röntgen-rays have been shown to accelerate 150 times the process of mutation. Muller, in his experiments with several thousand cultures of the fruit fly, has established the above ratio of induced mutations, which become hereditary.[11] 'Cosmic rays' in the form of radiation from the earth, in tunnels, for instance, show similar results, except that mutation occurs only twice as often as under the usual laboratory conditions. Under the influence of X-rays, mice change their colour of hair; gray mice become white, and white ones darker. Sometimes further additional bodily changes appear; as, for instance, one or no kidneys, abnormal eyes or legs, occur more often than under ordinary conditions. Some animals lose their power of reproduction, although the body is not obviously changed. Plants respond also to the X-ray treatment. They grow faster, flower more, and produce new forms more readily. In humans the effect of X-ray irradiation has often proven disastrous to the health of experimenters. There are even data that the irradiation of pregnant mothers may result in deformation of the head and limbs of the unborn child and, in one-third of the cases, feeble-mindedness of the children has resulted.[12]

Ultra-violet rays also show a marked effect. In some instances, they slow down or stop the streaming of protoplasm, because of increased viscosity or coagulation; plants grow slowly or rapidly; certain valuable ingredients in plants are increased; certain animals, as, for instance, small crustacea or bacteria are killed; eggs of Nereis (a kind of sea worm), which usually have 28 chromosomes, after irradiation have 70; certain bone malformations in children are cured; the toxin in the blood serum of pernicious anaemia patients is destroyed,.[13] In this respect, we should notice again that ultra-violet irradiation produces curative effects

like those of cod-liver oil, which shows that the effect of both factors is ultimately colloidal and structural.

Extensive experimentation with cathode rays is very recent, but already we have a most astonishing array of structural facts. Moist air is converted into nitric acid, synthetic rubber is produced rapidly, the milk from rubber trees is made solid and insoluble without the use of sulphur, liquid forms of bakelite are solidified without heating, linseed oil becomes dry to the touch in three hours and hard in six hours, certain materials, like cholesterol, yeast, starch, cottonseed oil, after exposure for thirty seconds, heal rickets, and similar unexpected results. What are usually called 'vitamins' do not only represent 'special substances', but become structurally active factors; and this is why ultra-violet rays may produce results like those of some 'substance'. It seems that in 'vitamins' the surface activities are important; the parallelism shown by von Hahn between the surface activities of different materials and the Funk table of vitamin content is quite suggestive. Some data seem to show that, in some instances, surface-active materials, such as coffee or alcohol, produce beneficial surface activities similar to the 'vitamins'.[14]

The above short list gives only an approximate picture of the overwhelming importance of the roles which structure in general, and colloids in particular, play in our lives. We see about us many human types. Some are delicate, some are heavy-set, some flabby, some puffy, all of which indicates a difference in their colloidal structure. Paired with these physical colloidal states are also nervous, 'mental', and other characteristics, which vary from weak and nervous to the extreme limitation of nervous activities, as in idiocy, which is a negation of activity.

It is curious that in all illnesses, whether 'physical' or 'mental', the symptoms are very few, and fundamentally of a standard type. In 'physical' illness we find the following common characteristics: fever, chills, headaches, convulsions, vomiting, diarrhoea,. In 'mental' ills, identifications, illusions, delusions, and hallucinations—in general, the reversed pathological order—are found. It is not difficult to understand the reason. Because of the general colloidal background of life, different disturbances of colloidal equilibrium *should* produce similar symptoms. In fact, many of these symptoms have been reproduced experimentally by injecting inert precipitates incapable of chemical reactions, which have induced artificial colloidal disturbances. Thus, if the serum from an epileptic patient is injected into a guinea pig, it results in an attack of convulsions, often ending in death. But, if the guinea pig is previously made immune by an injection of some colloid which accustoms the nerve-endings to the colloidal flocculation, then, for a few hours following, we

can, with impunity, introduce into the circulation otherwise fatal doses of epileptic serum. Epileptic serum can also be made immune by filtration, or by strong centrifugation, or by long standing, which frees it from colloidal precipitates.[15]

Death through blood transfusion or the injection of *any* colloid into the circulation has also, in the main, similar symptoms, regardless of the chemical character of the colloid, indicating once more the importance and fundamental character of structure.[16]

That illnesses are *somehow* connected with colloidal disturbances (note the wording of this statement) becomes quite obvious when we consider catarrhal diseases, inflammations, swellings, tumours, cancer, blood thrombi., which involve colloidal injuries, resulting in extreme cases in complete coagulation or fluidification, the variation between 'gel' and 'sol' appearing in a most diversified manner.[17] Other illnesses are connected with precipitation or deposits of various materials. Gout, for instance, results from a morbid deposit of uric acid, and different concretions, such as the 'stones', are very often found in different fluids of the organism. We have, thus, concretions in the intestines, the bile, the urine, the pancreas, the salivary glands; lime deposits in old softened tissues, 'rice bodies' in the joints, 'brain sand',.[18]

In bacterial diseases, the micro-organisms rapidly produce acids and bases which tend to destroy the colloidal equilibrium. Lately, it has been found that even tuberculosis is more than a mere chapter in bacteriology. All the main tubercular symptoms can be reproduced, experimentally, by means of colloidal disturbances without the intervention of a single bacterium.[19] This would explain also why, in some instances, psychotherapy is effective in diseases with tubercular symptoms.[20]

By structural necessity, every expression of cellular activity involves some sort of colloidal behaviour; and any factor disturbing the colloidal structure must be disturbing to the welfare of the organism. Vice versa, a factor which is beneficial to the organism must reach and affect the colloids.

After this brief account of the structural peculiarities of the domain in which life is found, we can understand the baffling 'body-mind' problem. We do not yet know as many details as we could wish, but these will accumulate the moment a general solution is clearly formulated. It is a well-established *experimental* fact that all nervous and 'mental' activities are connected with, or actually generate, electrical currents, which of late are scrupulously studied by the aid of an instrument called the psychogalvanometer.[21] It is not suggested that electrical currents are the only ones which are involved. There may be many different forms

of radiant energy produced or effective, which we have not yet the instruments to record. Experiments suggest such a possibility. Thus, for instance, the apex of a certain rapidly growing vegetable or animal tissue emits some sort of invisible radiation which stimulates the growth of living tissue with which it is not in contact. The tip of a turnip or onion root, if placed at right angles to another root, at a distance of a quarter of an inch, so stimulates the growth of the latter that the increase of the number of cells, on its side nearest the point of stimulation, is as high as seventy per cent. These radiations accelerate the growth of some bacteria. Other examples could be given.[22]

A classical example of the effect left on protoplasm by energetic factors is given by Bovie.[23]

As yet, we have not assumed that the protoplasm of plants also shows lasting structural and functional results of stimulation, some sort of 'learning' or 'habit-formation' characteristics. But such is the case; and further experimentation along these lines will help greatly to understand the mechanism of 'mental' processes in ourselves.

If we take the seed of a plant, for instance, of a squash, and keep it in a moist tropism chamber in the dark, it will grow a root. When the root is about one inch long, we begin our experiment. Originally, under the influence of gravitation, the root grows vertically downwards (A). If we rotate the tropism chamber 90° so that the root is horizontal (B), the root will soon bend downwards under the influence of positive geotropism. But the bending does not occur at once. There is a latent period—in the case of the squash seed, about ten minutes—after which pause the root is bent downwards. When we have determined this latent period for a given seedling, we then rotate the chamber to the positions (B), (C), (A), (B), (C)., just within the 'time' limit before the bending would occur. We repeat such procedure several times. When we set the root again in its vertical downward position (A), we notice that the root, without any more changes of position, will wag backwards and forward with the period as was used in the experiment. This unexpected behaviour will last for several days. It shows that the alternating stimulus of gravitation, as applied to the root, has produced some structural changes in the protoplasm which persist for a comparatively long period after the stimulus has ceased to act. It be-

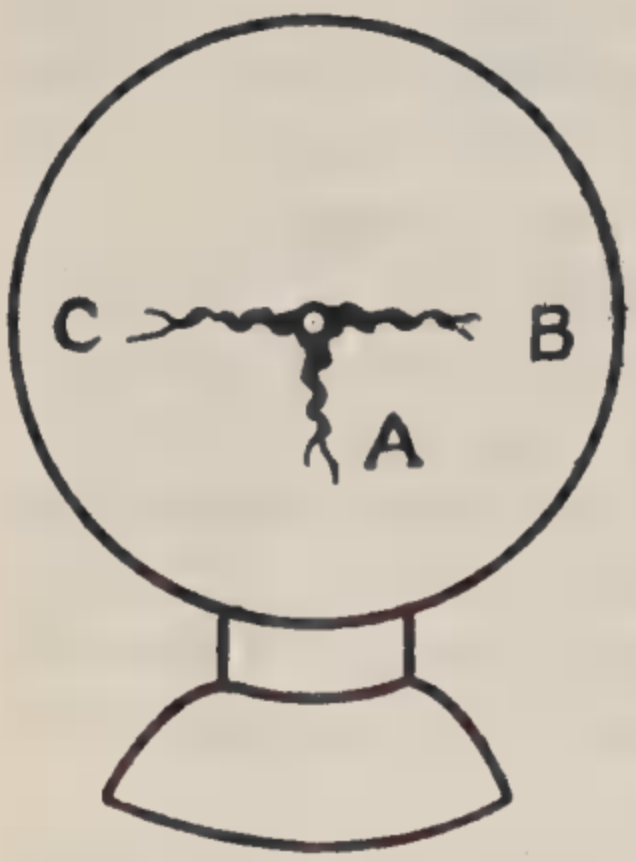

FIG. 1

comes obvious that teachability and the structural tendency for forming engrams is a general characteristic of protoplasm.

All the examples given above show clearly that *structure* in general, and of colloids in particular, gives us a satisfactory basis for the understanding of the *equivalence* between occurrences which belonged formerly to 'chemistry' and those classified as 'physical', and ultimately between these and those we call 'mental'. Structure, and structure alone, gives not only the *unique* content of what we call 'knowledge', but also the bridge between the different classes of occurrences—a fact which, as yet, has not been fully understood.

To sum up: It is known that colloidal behaviour is exhibited by materials of very fine subdivision, the 'world of neglected dimensions', which involves surface activities and electrical characters of manifold and complex structure, and therefore the flexibility of gross macroscopic characteristics. It is well known that all life-processes, 'feelings', 'emotions', 'thought', semantic reactions, and so forth, involve *at least* electrical currents. As electrical currents and other forms of energy are able to affect the colloidal structure on which our physical characteristics depend, obviously 'feelings', 'emotions', 'thought'.; in general, *s.r*, which are connected with manifestations of energy, will also have some effect on our bodies, and vice versa. Colloidal structure supplies us with an extremely flexible mechanism with endless possibilities.

When we analyse the known empirical facts from a structural point of view, we find not only the equivalence which was mentioned before, but we must, also, legitimately consider the so-called 'mental', 'emotional', and other semantic and nervous occurrences in connection with manifestations of energy which have a powerful influence on the colloidal behaviour, and so ultimately on the behaviour of our organisms as-a-whole. Under such environmental conditions, we must take into account all energies which have been discovered, *semantic reactions not excluded,* as all such energies have structural effect. As language is one of the expressions of one of these energies, we ought to find it quite natural that the structure of language finds its reflection in the structure of the environmental conditions which are dependent on it.

Until lately, the disregard of colloidal science and of structure in general has greatly retarded advance in biology, psychiatry, and other sciences. Biology, for instance, has mostly studied 'life' where none existed; namely, in death. If we study corpses, we study death, not life, and life is a function of living cells. The living cell is semi-fluid, and all the forces which act in colloidal solutions and constitute colloidal

behaviour are acting because they *can act,* while a dead cell is *coagulated* and so a different set of energies is operating there.[24]

Should we wonder that life, being a form of colloidal behaviour on microscopic and sub-microscopic levels, conditioned by little colloidal 'wholes', and structures separated from their environment by surfaces, preserves a similar character on macroscopic levels? We should, instead, be surprised if this did not turn out to be the case.